

SELVAKUMAR SADURSHAN

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Summary

Final-year Data Science undergraduate at SLIIT working across the stack and machine learning. I've worked on production-grade backend systems using .NET, Typescript and I've also focused on model optimization and evaluation. One of my projects was a predictive-maintenance framework tested on NASA C-MAPSS data, where I built and evaluated models to predict equipment failures. I've also worked on edge-deployed segmentation models optimizing them to run in under 50 milliseconds on CPU. I'm comfortable taking ownership of a feature from start to finish. From the data pipeline through the API all the way, to the deployed user interface.

Professional Experience

Intern - Data analyst, HSBC

Aug 2025 – Feb 2026

- Performed data extraction, cleaning, validation, and customer data matching using Python and Excel.
- Developed Power BI/Excel dashboards and reports to analyze department performance and work quality.
- Automated email workflows and customer identification processes using Excel Macros and Python, improving operational efficiency.

Student Counselor PrimeleedManagement

- Served as the main point of contact for UK clients and universities, ensuring clear communication and strong relationships.
- Guided students on course selection, university admissions, and application processes.

Jun 2022 – Oct 2022

Course-Consultant Officer Eclub

- Conducted workshops to explain e-club courses in an interactive way.
- Helped students gain practical, hands-on skills through these sessions.

Aug 2021 – Feb 2022

Education

Bachelor's Degree in Data-Science, SLIIT

2022 feb – Present

Certificate level of BCS Higher Education Qualification (HEQ) *Live College*
BCS, The Chartered Institute for IT : 1st Stage Completed

2023 mar - 2023 Sep

Diploma In IOT *Esoft Metro Campus*
Basics Of IOT

2019 jan - 2019 june

Skills

Programming Languages

R, C, Java, Python, C++, kotlin, React-Native, Java-Script, ASP.NET Core

Mobile Development

Android Application Development

Data-warehousing

ETL, SQL, Data cleaning, Data Transformation

Allibraries

Tensor-flow, Sklearn, kears, Pytorch

Web Development

MERN Stack, Spring-Boot, django

UI/UX Design

UI Designing, Figma, Draw.io

Databases:

Superbase, PostgreSQL, Firebase, MongoDB

Tools

Power-BI, Trello, Kanban board, Docker

Certifications

- | | | |
|---|--|---|
| ■ Introduction to Party Rock :
AWS Educate | ■ Prompt Engineering : Greeting
Learning | ■ Introduction to Generative Ai :
AWS Educate |
| ■ IBM data-Analyst : Udemy | ■ IBM Machine Learning With
Python : Coursera | ■ Advanced ML Algorithms &
Unsupervised Learning : Packt |

Projects

Anomaly and Health Monitoring System On Turbo fan engine of aircraft

- Developed an Explainable AI-based health monitoring and anomaly detection system using NASA C-MAPSS turbofan data.
- Implemented LSTM digital twin and ensemble anomaly detection using GMM, Isolation Forest, LSTM autoencoder, and K-Means.
- Performed time-series processing, feature engineering, root-cause analysis, and anomaly visualisation using Python.

Edge Defect Segmentation

- Developed an industrial surface-defect segmentation model using KolektorSDD2, targeting edge deployment under a 50 ms P95 CPU latency constraint.
- Applied knowledge distillation, structured pruning, INT8 quantization, and mixed-precision optimization to reduce model cost while preserving segmentation performance.
- Conducted multi-configuration CPU benchmarking, per-layer sensitivity analysis, and quantization ablation to identify the optimal deployment configuration.

hybrid-rag-assistant

- Built a RAG-based document assistant using FAISS, BM25, MiniLM embeddings, and cross-encoder reranking.
- Implemented grounded AI responses with page-level citations, conversational memory, and hallucination/refusal control.
- Developed an end-to-end FastAPI + React application with document upload, retrieval, streaming, and structured extraction.
- Achieved 1.00 Hit@1 and 1.00 MRR through retrieval evaluation and optimization.

Workshop Management

- Built a full-stack vehicle service management system for job cards, invoicing, and workflow management using Next.js 15, TypeScript, PostgreSQL, Prisma, and Auth.js.
- Implemented a layered architecture with services and repositories, keeping business logic independent of HTTP handlers and achieving 330+ unit/integration tests plus Playwright E2E testing.
- Developed transaction-safe job-status workflows, audit trails, and gapless invoice numbering, with concurrency testing for reliable financial operations.
- Implemented role-based authorization and precise Decimal-based currency handling, preventing unauthorized job-card access and floating-point errors in

BusTiker — Bus Seat Reservation System

- Engineered a concurrency-safe seat reservation engine for a Jakarta EE booking platform
- Remediated 9 classes of vulnerability in a legacy servlet codebase — SQL injection, plaintext passwords, IDOR, CSRF, XSS and session fixation
- Rebuilt the system on a strict layered architecture (servlet → service → DAO → model) with HikariCP connection pooling

AppointFlow

- Built a multi-tenant SaaS platform with Clean Architecture and CQRS across four layered assemblies
- Enforced tenant data isolation via EF Core global query filters, verified by integration tests covering all cross-tenant access paths
- Prevented double-booking using optimistic concurrency and database constraints, validated under 20 concurrent booking attempts